Mutual Information

Definition and Properties

IEOR E4576 — Data-Driven Models in Finance — Fall 2026 — Week 4

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1 Definition

The **mutual information** between two discrete random variables X and Y is

$$I(X;Y) = \sum_{x,y} p(x,y) \log \frac{p(x,y)}{p(x)p(y)}.$$

For continuous random variables with joint density $f(x,y)$ and marginals $f(x)$ and $f(y)$ the sum becomes an integral:

$$I(X;Y) = \iint f(x,y) \log \frac{f(x,y)}{f(x)f(y)} \mathrm{d}x\mathrm{d}y.$$

Mutual information quantifies how much knowing one variable reduces uncertainty about the other. It is equivalently the Kullback–Leibler divergence between the joint distribution and the product of the marginals,

$$I(X;Y) = D_{\mathrm{KL}}\big(p(x,y) \| p(x)p(y)\big),$$

and it can be written in terms of entropies:

$$I(X;Y) = H(X) - H(X \mid Y) = H(Y) - H(Y \mid X) = H(X) + H(Y) - H(X,Y).$$

Mutual information measures any dependence, not just linear dependence. It is zero exactly when X and Y are independent and positive otherwise, so it detects nonlinear and non-monotonic relationships that a correlation coefficient misses. The price is that it carries no sign: it says *how much* one variable tells us about the other, not in which direction.

2 Key Properties

Non-negativity
$I(X;Y) \geq 0$, with equality if and only if X and Y are independent.

Symmetry
$I(X;Y) = I(Y;X)$.

Captures nonlinear dependence
Unlike Pearson correlation, mutual information detects any form of statistical dependence, including nonlinear and non-monotonic relationships.

Data processing inequality
If $X \to Y \to Z$ is a Markov chain, then $I(X;Z) \leq I(X;Y)$: processing Y into Z cannot increase the information it carries about X.

Invariance under bijections
If ϕ and ψ are invertible, then $I(\phi(X);\psi(Y)) = I(X;Y)$.

3 Normalised Mutual Information

Raw mutual information has no fixed scale: its magnitude depends on the entropies of X and Y, which makes it hard to interpret in isolation. The **normalised mutual information** (NMI) rescales it to $[0, 1]$, where 0 indicates independence. The normalisations in common use differ only in the entropy term in the denominator:

Normalisation	Behaviour
$I(X;Y)/\sqrt{H(X)H(Y)}$	Geometric mean. Equals 1 when each variable fully determines the other.
$I(X;Y)/\min\{H(X), H(Y)\}$	Reaches 1 as soon as either variable determines the other.
$I(X;Y)/\frac{1}{2}\big(H(X) + H(Y)\big)$	Arithmetic mean; the default in scikit-learn.